Indicators of fire – *Pteris* (fern) and *Pyronema* (fungus) indicates the burnt up and fire disturbed areas. So they are called indicators of fire.

Fire break – It is a gap made in the vegetation that acts as a barrier to slow down or stop the progress of fire.

A **natural fire break** may occur when there is a lack of vegetation such as River, lake and canyon found in between vegetation may act as a natural fire break.

Rhytidome: It is the structural defense by plants against fire .The outer bark of trees which extends to the last formed periderm is called Rhytidome. It is composed of multiple layers of suberized periderm, cortical and phloem tissues. It protects the stem against fire , water loss, invasion of insects and prevents infections by microorganisms.

6.2.2 Edaphic factors

Edaphic factors, the abiotic factors related to soil, include the physical and chemical composition of the soil formed in a particular area. The study of soils is called **Pedology.**

The soil

Soil is the weathered superficial layer of the Earth in which plants can grow. It is a complex composite mass consisting of soil constituents, soil water, soil air and soil organisms, etc.

Soil formation

Soil originates from rocks and develops gradually at different rates, depending upon the ecological and climatic conditions. Soil formation is initiated by the weathering process. Biological weathering takes place when organisms like bacteria, fungi, lichens and plants help in the breakdown of rocks through the production of acids and certain chemical substances.

Soil types

Based on soil formation (**pedogenesis**), the soils are divided into

1. **Residual soils** –These are soils formed by weathering and **pedogenesis** of the rock.
2. **Transported soils** – These are transported by various agencies.

The important edaphic factors which affect vegetation are as follows:

1. Soil moisture: Plants absorbs rain water and moisture directly from the air

2. Soil water: Soil water is more important than any other ecological factors affecting the distribution of plants. Rain is the main source of soil water. Capillary water held between pore spaces of soil particles and angles between them is the most important form of water available to the plants.

3. Soil reactions: Soil may be **acidic** or **alkaline** or **neutral** in their reaction. pH value of the soil solution determines the availability of plant nutrients. The best pH range of the soil for cultivation of crop plants is **5.5 to 6.8.**

4. Soil nutrients: Soil fertility and productivity is the ability of soil to provide all essential plant nutrients such as minerals and organic nutrients in the form of ions.

5. Soil temperature: Soil temperature of an area plays an important role in determining the geographical distribution of plants. Low temperature reduces use of water and solute absorption by roots.

6. Soil atmosphere: The spaces left between soil particles are called pore spaces which contains **oxygen** and **carbon-di-oxide**.

7. Soil organisms: Many organisms existing in the soil like bacteria, fungi, algae, protozoans, nematodes, insects, earthworms, etc. are called soil organisms.

IR2PF7

TN_GOVT_BIOLOGY-BOTANY_XII_PAGES_122-147 CH 06.indd 115 3/14/2020 11:38:35 AM